

EXHIBIT 21

Ex. 21 - Infringement Contention Claim Chart
U.S. Patent No. 12,298,394 (“the ’394 patent”)

Apple’s products directly and/or indirectly infringe at least claims 1, 5, 6, 15, 18, 19 (“Asserted Claims”) of U.S. Patent No. 12,298,394 (“the ’394 patent”). The products that infringe the ’394 patent include at least the iPhone 15, iPhone 15 Plus, iPhone 15 Pro, iPhone 15 Pro Max, iPhone 16, iPhone 16e, iPhone 16 Pro, iPhone 16 Pro Max, iPhone 16 Plus, iPhone 17, iPhone 17e, iPhone 17 Pro, iPhone 17 Pro Max, iPhone Air, 11-inch iPad Pro (4th Generation), iPad Pro 12.9 inch (6th Generation), iPad Pro 11-inch (M4 and M5), iPad Pro 13-inch (M4 and M5) and other infringing iPhone and iPad models (each an “Accused Apple Product” and collectively the “Accused Apple Products”). The Accused Apple Products infringe the ’394 patent as set forth below. Each of the following limitations is met literally, and, to the extent a limitation is not met literally, it is met under the doctrine of equivalents.

Additionally, the Apple iPhone 17 is representative of the Accused Apple Products at least because the other Accused Apple Products function in the same or substantially similar manner to the Apple iPhone 17.¹ The relevant identification processes and functionalities, as well as the hardware components of the Apple iPhone 17, are demonstrated through testing of the Apple iPhone 17 (“Apple Product Testing”), which is available for inspection upon request.

Claim 1	Accused Apple Products
[1pre] A system, comprising:	Each Accused Apple Product constitutes a system. For example, the Apple iPhone 17 is a system:

¹ Each of the Accused Apple Products includes the claimed hardware components and steps. The hardware components and source code implementing the relevant functionalities of the Apple iPhone 17 are representative of the Accused Apple Products for the purpose of infringement. To the extent there may be differences in the hardware composition, certain software steps, or source code versions among these Accused Apple Products, those differences are not material for the infringement analysis.

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Exhibit 36.

[1a] a structured light projector configured to emit a plurality of

The Accused Apple Products contain a structured light projector configured to emit a plurality of light beams, arranged in a first spatial arrangement relative to each other, towards an object.

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light beams, arranged in a first spatial arrangement relative to each other, towards an object;

For example, the Apple iPhone 17 includes a TrueDepth camera system that enables features like FaceID.

Face ID

Enabled by TrueDepth technology in the Center Stage front camera

Exhibit 33.

APPLE IPHONE 17 DISASSEMBLY



Exhibit 34C, at 23.

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The TrueDepth camera system includes a dot projector Vertical-Cavity Surface-Emitting Laser (“VCSEL”), a flood illumination VCSEL, and a near-infrared camera module.

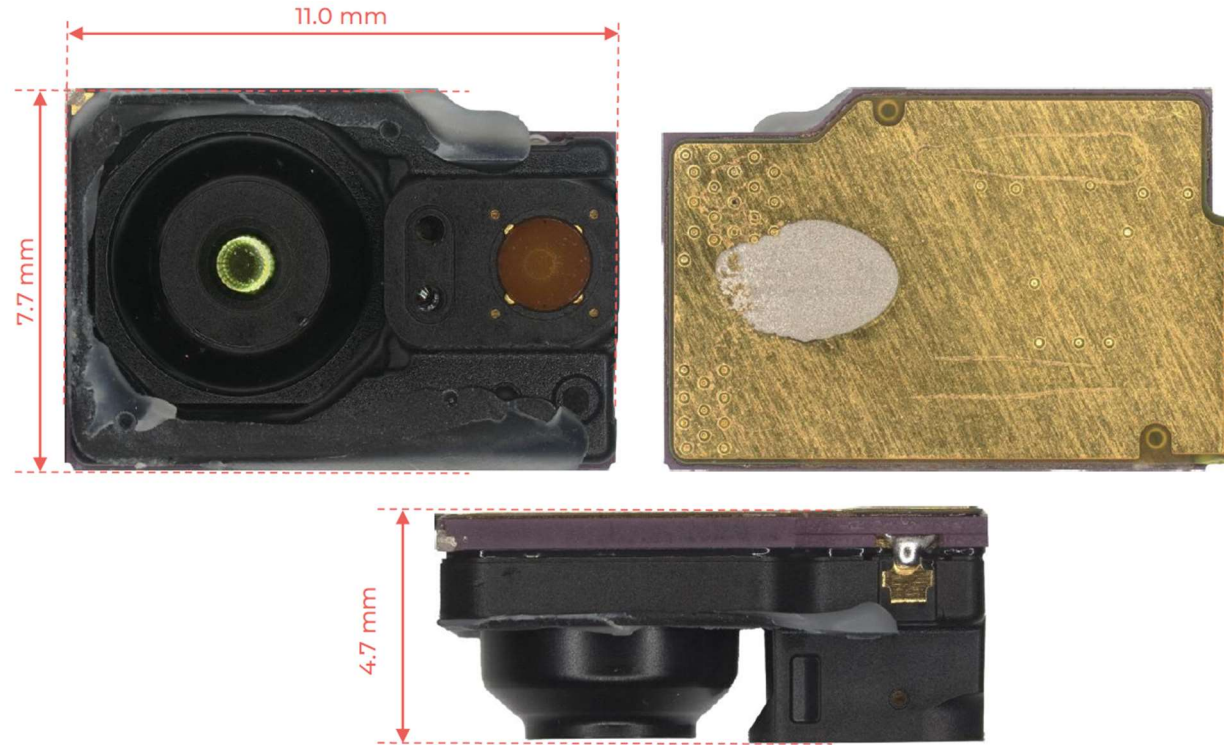


Exhibit 34C, at 35.

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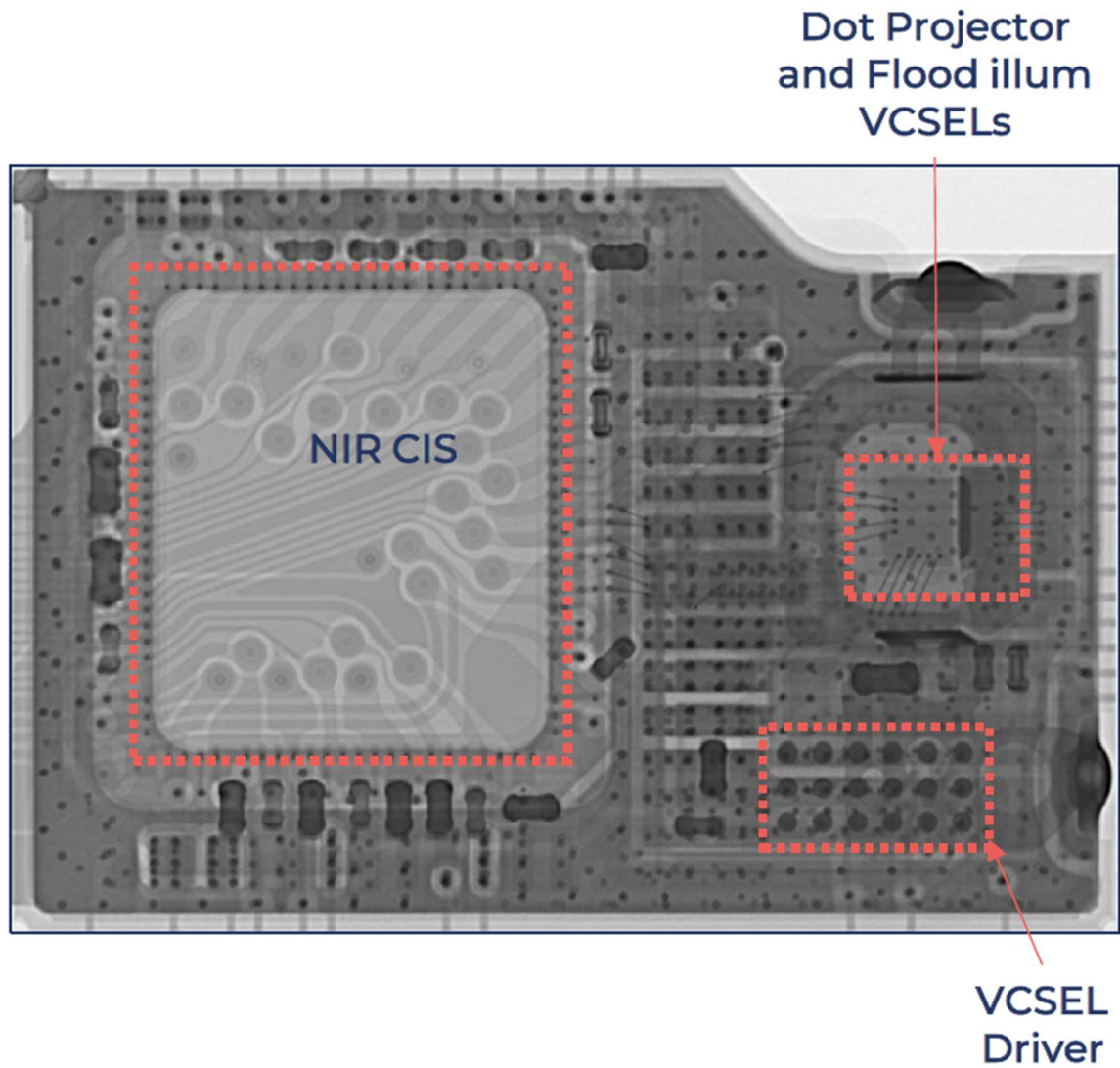


Exhibit 34C, at 36.

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iPhone 17 Pro Face ID
Module Opened © Yole
Group 2026

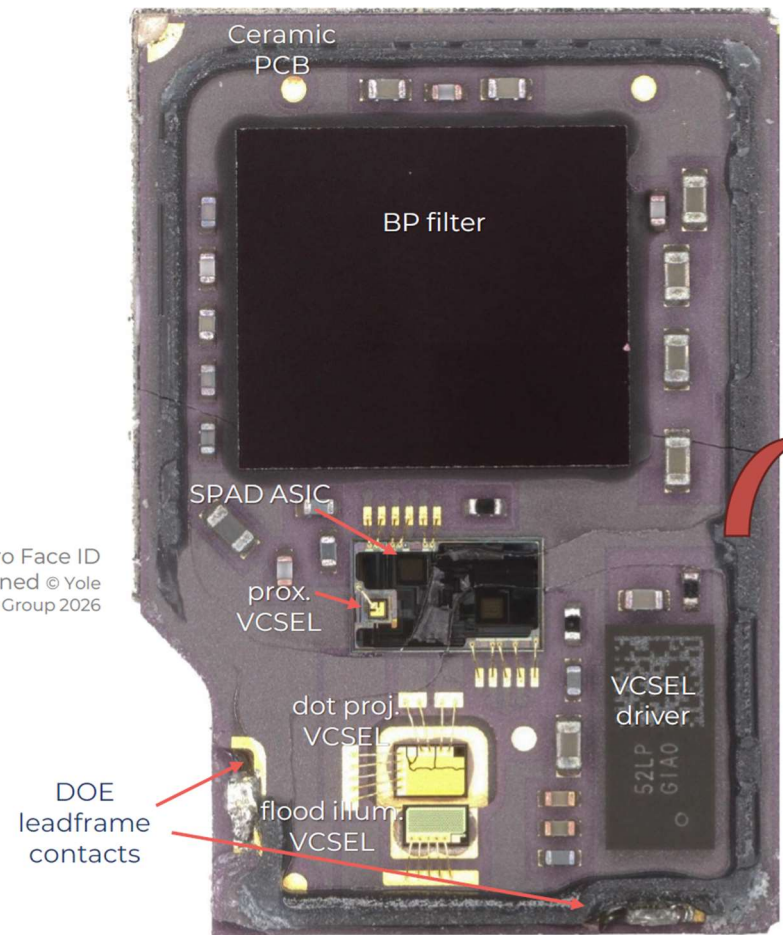


Exhibit 34C, at 37.

The TrueDepth camera system emits a plurality of light beams toward an object.

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	After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses. Exhibit 35, at 22. The TrueDepth camera system is configured to emit light beams arranged in a first spatial arrangement relative to each other. For example, the dot projection VCSEL employs a randomized arrangement.
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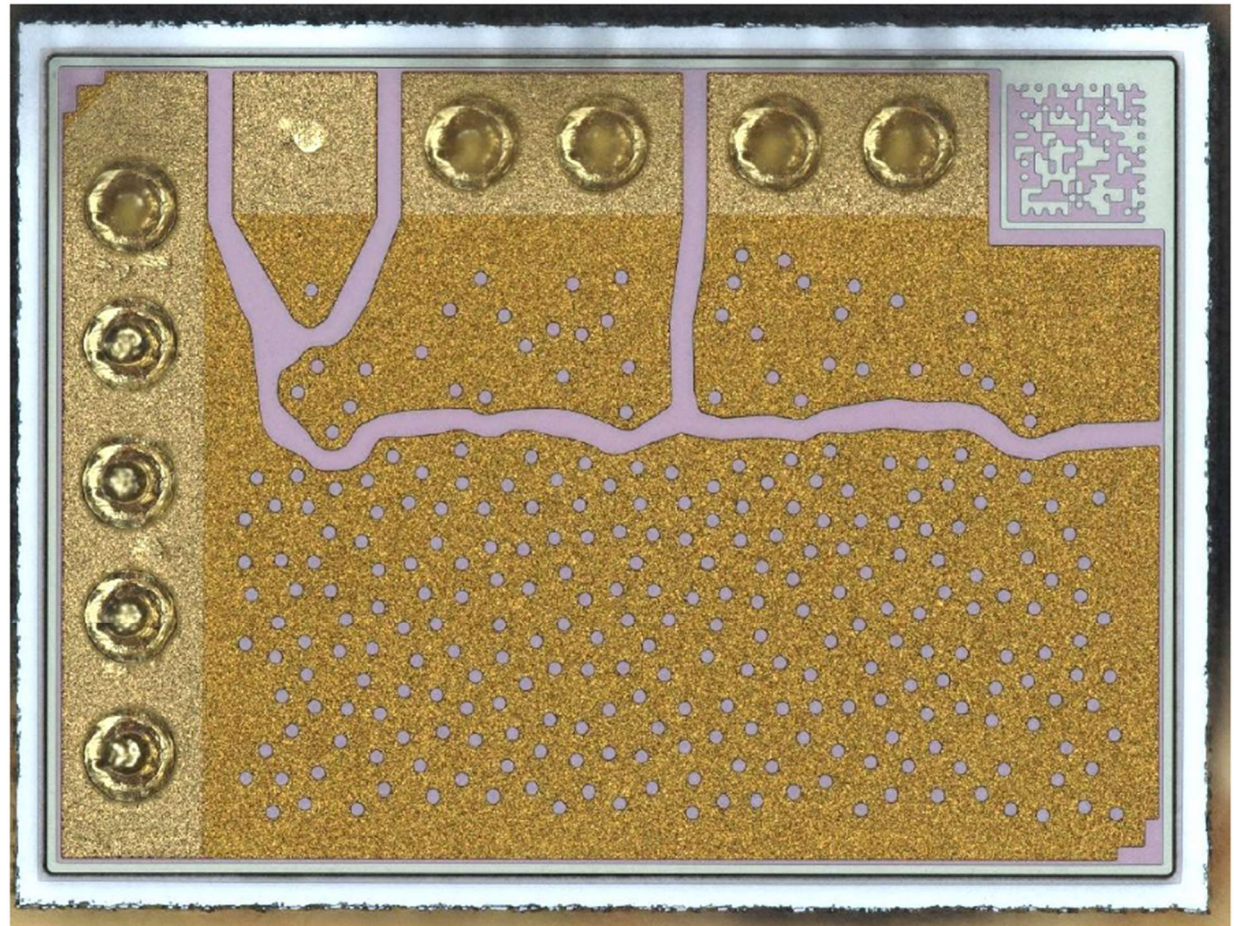


Exhibit 34C, at 77.

The flood arrangement VCSEL uses a regularized arrangement.

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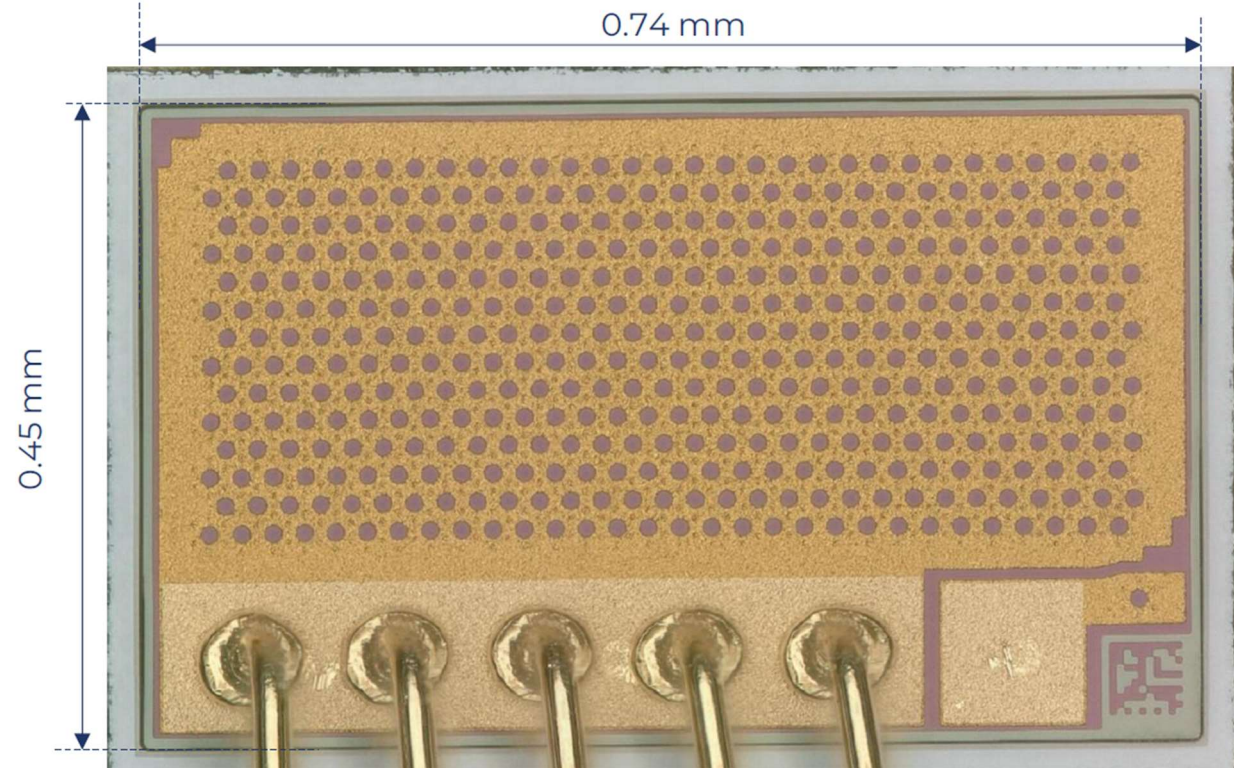


Exhibit 34C, at 86.

[1b] a semiconductor detector configured to receive a light reflective from the object and output an image characterizing an intensity of the reflective light, the semiconductor detector sensitive to light;

The Accused Apple Products contain a semiconductor detector configured to receive a light reflective from the object and output an image characterizing an intensity of the reflective light, the semiconductor detector sensitive to light.

For example, the Apple iPhone 17 includes a receiver module that receives light reflected from the object.

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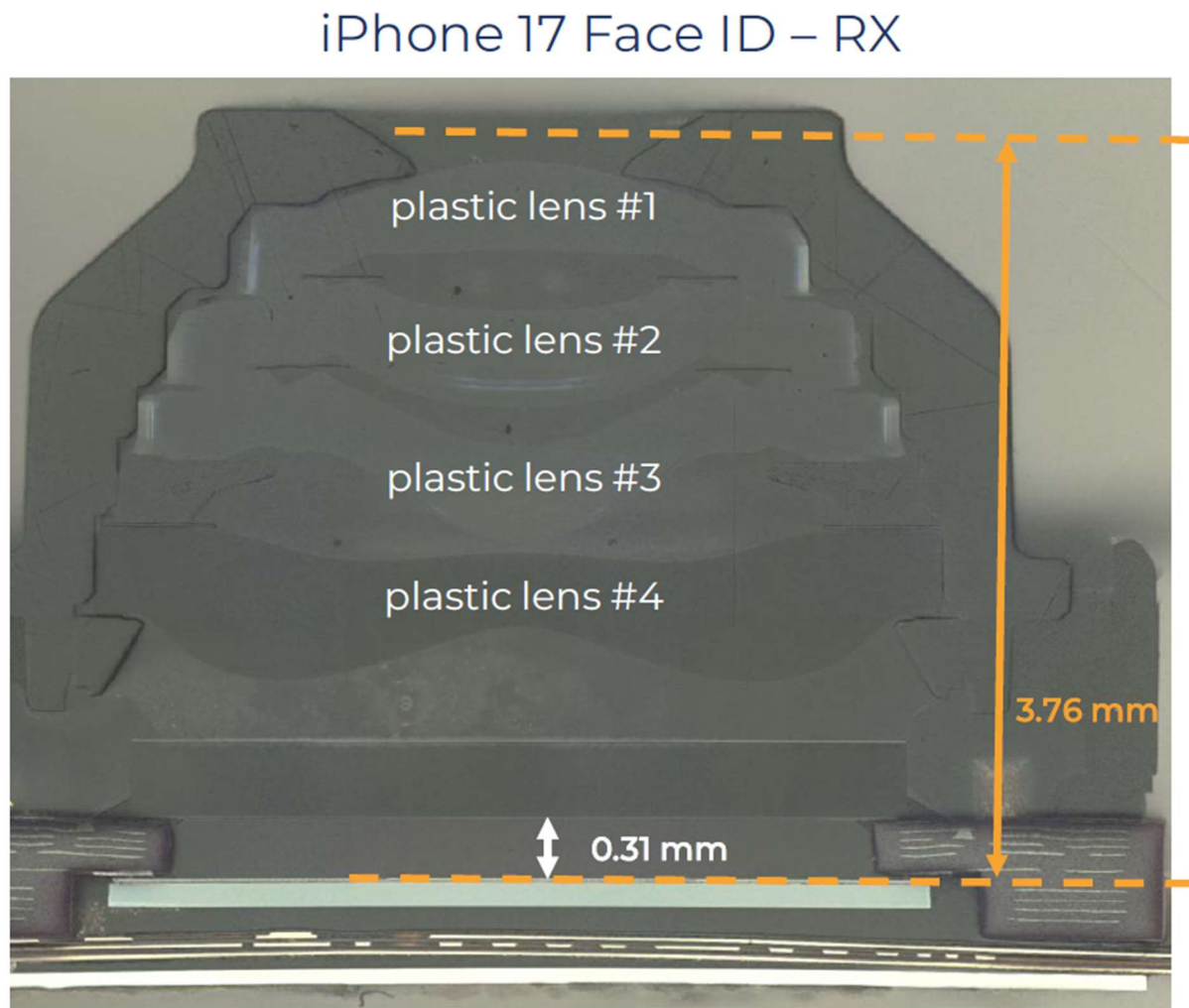


Exhibit 34C, at 27.

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The receiver module contains a Complementary Metal Oxide Semiconductor (“CMOS”) sensor that is sensitive to light and captures light reflected from the object to form an image.

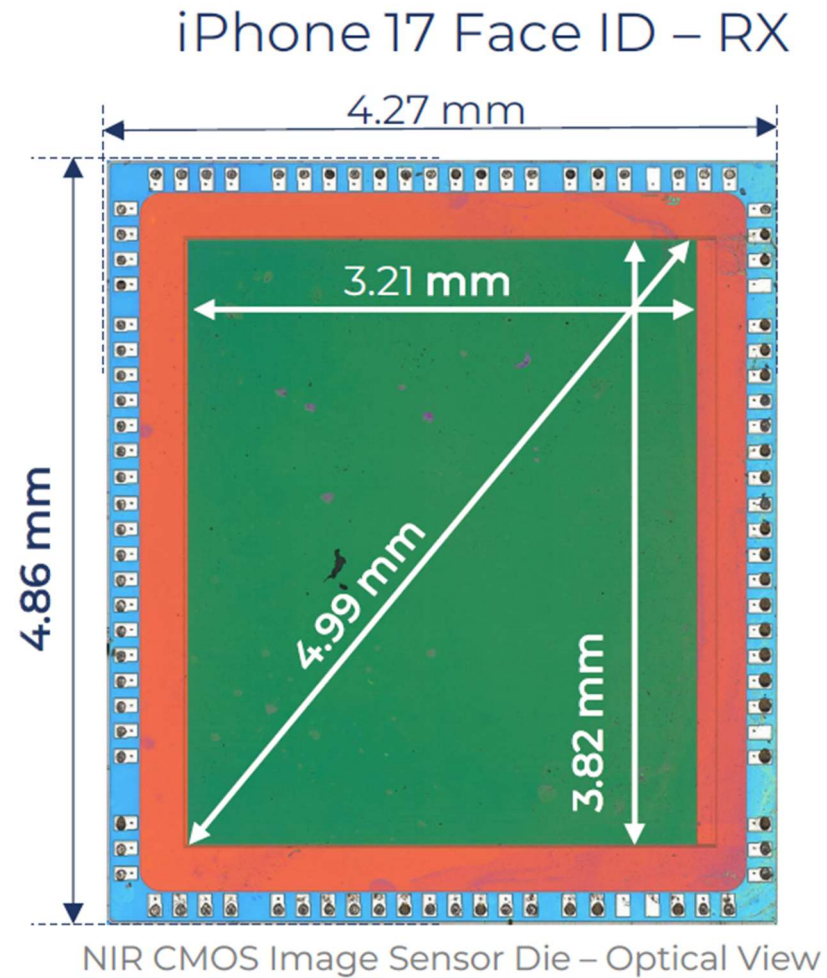


Exhibit 34C, at 28.

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	The receiver module produces an image characterizing an intensity of the reflective light. Apple’s biometric security architecture relies on a strict separation of responsibilities between the biometric sensor and the Secure Enclave, and a secure connection between the two. The sensor captures the biometric image and securely transmits it to the Secure Enclave. During enrollment, the Secure Enclave processes, encrypts, and stores the corresponding Optic ID, Face ID, and Touch ID template data. During matching, the Secure Enclave compares incoming data from the biometric sensor against the stored templates to determine whether to unlock the device or respond that a match is valid (for Apple Pay, in-app, and other uses of Optic ID, Face ID, and Touch ID). The architecture supports devices that include both the sensor and Secure Enclave (such as iPhone, iPad, many Mac computers, and Apple Vision Pro), as well as the ability to physically separate the sensor into a peripheral that’s then securely paired to the Secure Enclave in a Mac with Apple silicon. Exhibit 35, at 20. After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses. Exhibit 35, at 22.
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	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device's neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36.
[1c] a processor configured to:	The Accused Apple Products contain a processor. For example, the Apple iPhone 17 includes a processing chip. Chip A19 A19 chip 6-core CPU with 2 performance and 4 efficiency cores 5-core GPU with Neural Accelerators 16-core Neural Engine Hardware-accelerated ray tracing Exhibit 33.

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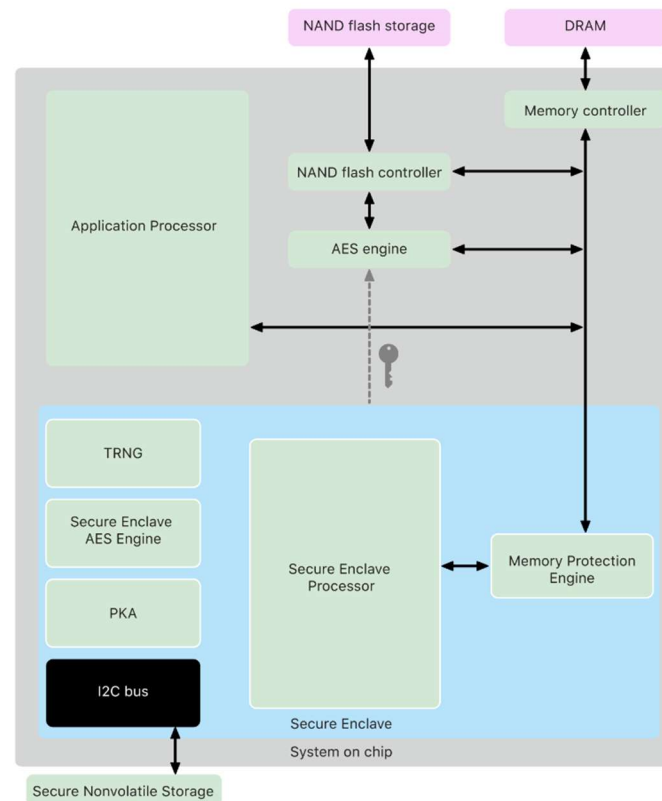
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Secure Enclave

The Secure Enclave is a dedicated secure subsystem in the latest versions of iPhone, iPad, Mac, Apple TV, Apple Watch, Apple Vision Pro, and HomePod.

Overview

The Secure Enclave is a dedicated secure subsystem integrated into Apple [system on a chip \(SoC\)](#). The Secure Enclave is isolated from the main processor to provide an extra layer of security and is designed to keep sensitive user data secure even when the Application Processor kernel becomes compromised. It follows the same design principles as the SoC does—a boot ROM to establish a hardware root of trust, an AES engine for efficient and secure cryptographic operations, and protected memory. Although the Secure Enclave doesn’t include storage, it has a mechanism to store information securely on attached storage separate from the NAND flash storage that’s used by the Application Processor and operating system.



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	Exhibit 35, at 10.
[1d] select at least one region of the image that characterizes the reflective light;	The Accused Apple Products contain a processor configured to select at least one region of the image that characterizes the reflective light. For example, Face ID on the Apple iPhone 17 analyzes a region of the image containing the face or the eyes that characterizes the reflective light. To start using Face ID, you need to first enroll your face. You might do this during the initial set up process, or at a later time by going to Settings > Face ID & Passcode. To unlock your device using Face ID, simply glance at it. Face ID requires that the TrueDepth camera sees your face or your eyes, whether your device is lying on a surface or you're holding it in a natural position. The TrueDepth camera has a similar range of view as when you take a photo or make a FaceTime call with the front camera. Face ID works best when the device is arm's length or less from your face (25–50 cm away from your face). To use Face ID while wearing a mask, set up the feature and make sure the camera can see your eyes. Exhibit 36.

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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

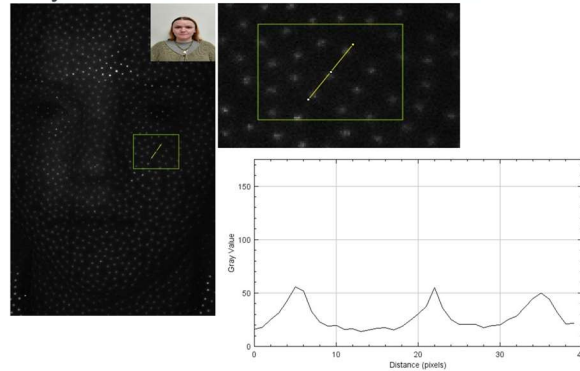
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[1e] determine a distribution of light intensity of a first subregion inclusive of a geometric center of the region;

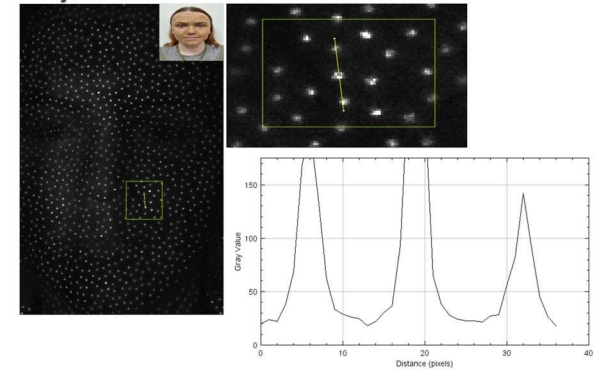
The Accused Apple Products contain a processor configured to determine a distribution of light intensity of a first subregion inclusive of a geometric center of the region.

For example, Face ID on the Apple iPhone 17 succeeds when given certain distributions of light intensity of a first subregion inclusive of the geometric center of the region, where the region constitutes the area around a dot, but fails when given other distributions. The different distributions constitute different shapes and brightnesses.

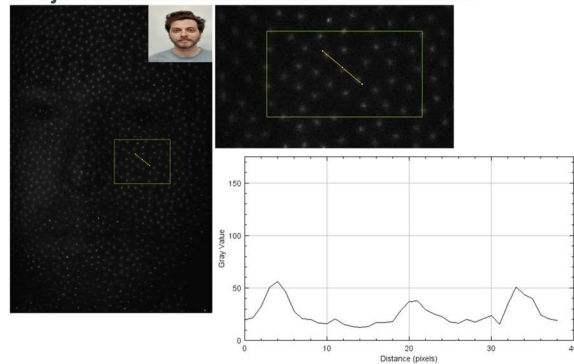
Subject01 Covered area: none – natural Face



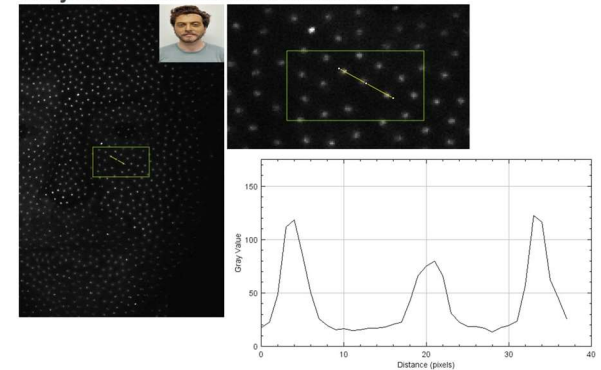
Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face



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	Exhibit 47 ¶¶ 23-31. Additionally and alternatively, Face ID works “even with a mask on,” indicating that Face ID determines distributions of particular facial regions.
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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

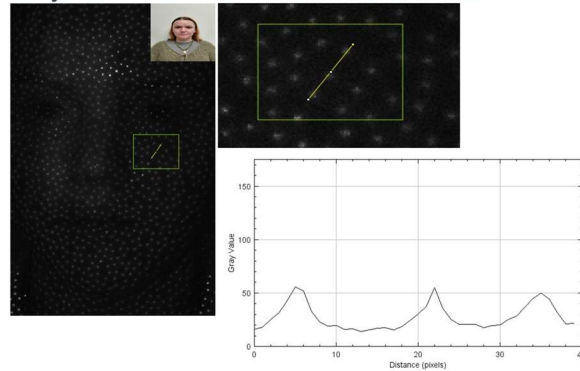
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[1f] determine a distribution of light intensity of a second subregion extending from the first subregion;

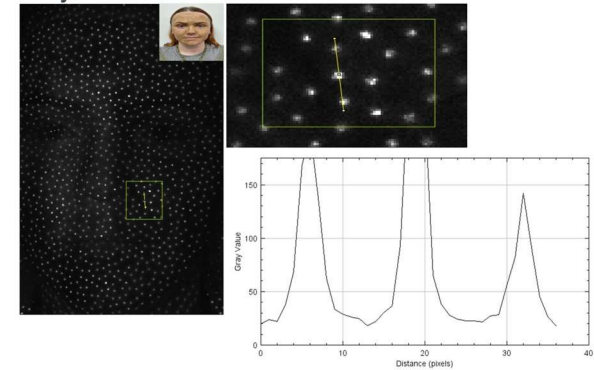
The Accused Apple Products contain a processor configured to determine a distribution of light intensity of a second subregion extending from the first subregion.

For example, Face ID on the Apple iPhone 17 succeeds when given certain distributions of light intensity of a second subregion extending from the first subregion, where the region constitutes the area around a dot, but fails when given other distributions. The different distributions constitute different shapes and brightnesses.

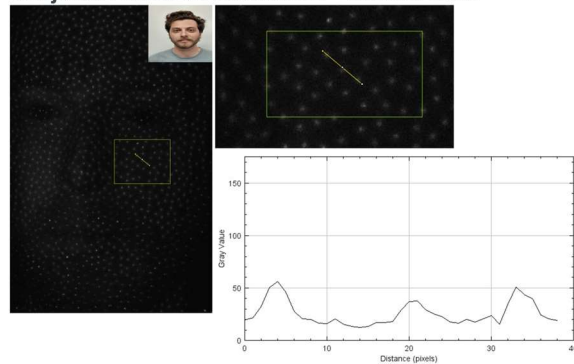
Subject01 Covered area: none – natural Face



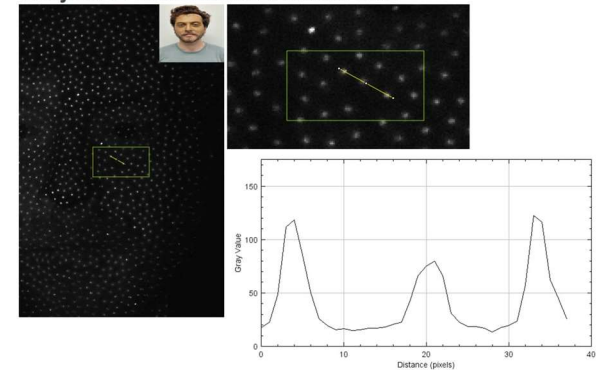
Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face



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	Exhibit 47 ¶¶ 23-31. Additionally and alternatively, Face ID works “even with a mask on,” indicating that Face ID determines distributions of particular facial regions.
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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

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[1g] compare the distribution of light intensity of the first subregion and the second subregion to a distribution of light intensity of a corresponding region in reference image data derived from a plurality reference images;	The Accused Apple Products contain a processor configured to compare the distribution of light intensity of the first subregion and the second subregion to a distribution of light intensity of a corresponding region in reference image data derived from a plurality reference images. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispooofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22. “Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.
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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

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Biometric security

Passcodes and passwords are essential to the security of Apple devices. At the same time, users require convenient access to their devices, often more than a hundred times a day. Biometric authentication provides a way to retain the security of a strong passcode—or even strengthen the passcode or password because it need not be entered manually—while providing the convenience of swiftly unlocking with a finger press or glance. Optic ID, Face ID, and Touch ID don’t replace a passcode or password, but in most situations they do make access faster and easier.

Apple’s biometric security architecture relies on a strict separation of responsibilities between the biometric sensor and the Secure Enclave, and a secure connection between the two. The sensor captures the biometric image and securely transmits it to the Secure Enclave. During enrollment, the Secure Enclave processes, encrypts, and stores the corresponding Optic ID, Face ID, and Touch ID template data. During matching, the Secure Enclave compares incoming data from the biometric sensor against the stored templates to determine whether to unlock the device or respond that a match is valid (for Apple Pay, in-app, and other uses of Optic ID, Face ID, and Touch ID). The architecture supports devices that include both the sensor and Secure Enclave (such as iPhone, iPad, many Mac computers, and Apple Vision Pro), as well as the ability to physically separate the sensor into a peripheral that’s then securely paired to the Secure Enclave in a Mac with Apple silicon.

Exhibit 35, at 20.

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Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

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Facial matching security

Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later. Furthermore, it’s designed to work indoors, outdoors, and even in total darkness. An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks. Face ID data, including mathematical representations of a user’s face, is encrypted and available only to the Secure Enclave. This data never leaves the device. It’s not sent to Apple, nor is it included in device backups. The following Face ID data is saved, encrypted only for use by the Secure Enclave, during normal operation:

- The mathematical representations of a user’s face calculated during enrollment
- The mathematical representations of a user’s face calculated during some unlock attempts if Face ID deems them useful to augment future matching

Face images captured during normal operation aren’t saved but are instead immediately discarded after the mathematical representation is calculated for either enrollment in Face ID or comparison with enrolled Face ID data.

Exhibit 35, at 29.

Ex. 21 - Infringement Contention Claim Chart
U.S. Patent No. 12,298,394 (“the ’394 patent”)

	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device’s neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36. Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It’s designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping). Exhibit 36.
[1h] determine a classification of the object based on the distribution of light intensity comparison of the first subregion and the second subregion to the corresponding region in the reference image data; and	The Accused Apple Products contain a processor configured to determine a classification of the object based on the distribution of light intensity comparison of the first subregion and the second subregion to the corresponding region in the reference image data. For example, Face ID on the Apple iPhone 17 classifies objects in the scene based on whether the object is an individual registered in Face ID for the given phone.

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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

Ex. 21 - Infringement Contention Claim Chart
U.S. Patent No. 12,298,394 (“the ’394 patent”)

About Face ID advanced technology

Learn how Face ID helps protect your information on your iPhone and iPad Pro.

Much of our digital lives are stored on iPhone and iPad, and it's important to protect that information. In the same way that Touch ID revolutionized authentication using a fingerprint, Face ID revolutionizes authentication using facial recognition. Face ID provides intuitive and secure authentication enabled by the state-of-the-art TrueDepth camera system with advanced technologies to accurately map the geometry of your face.

With a simple glance, Face ID securely unlocks your iPhone or iPad Pro. You can use it to authorize payments with Apple Pay, purchases from the App Store, Book Store, and other Apple media services, and more. Developers can also allow you to use Face ID to sign into their apps.

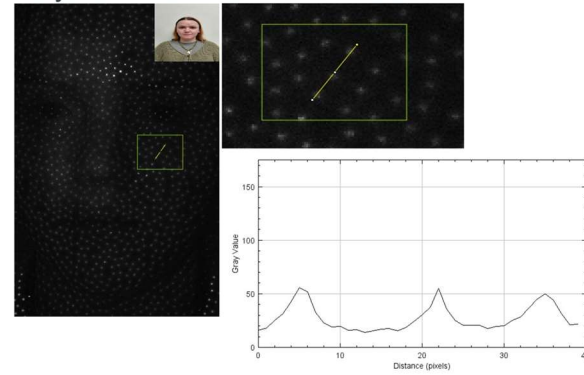
Exhibit 36.

This determination is based on a light intensity comparison of the first subregion and the second subregion to the corresponding region in the reference image data.

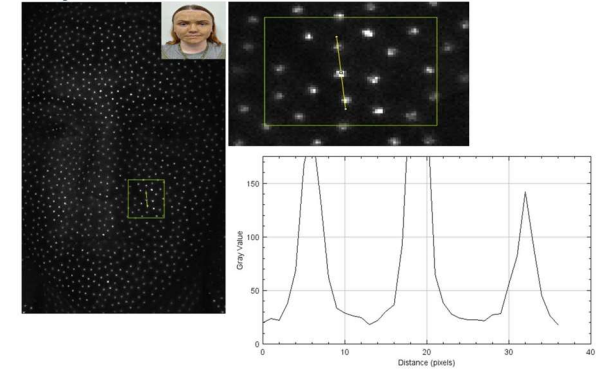
For example, Face ID on the Apple iPhone 17 succeeds when given certain distributions of light intensity, but fails when given other distributions.

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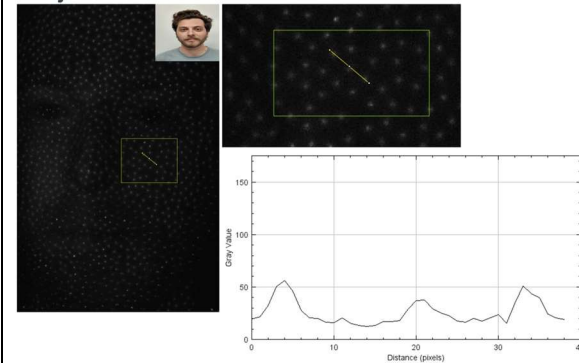
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

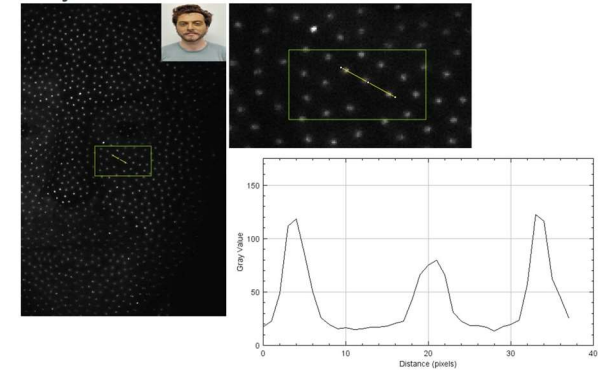


Exhibit 47 ¶¶ 23-31.

Ex. 21 - Infringement Contention Claim Chart
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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

Ex. 21 - Infringement Contention Claim Chart
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	Face ID also uses neural networks to compare the given regions to corresponding regions in reference image data.
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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispooofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

Ex. 21 - Infringement Contention Claim Chart
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Facial matching security

Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later. Furthermore, it’s designed to work indoors, outdoors, and even in total darkness. An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks. Face ID data, including mathematical representations of a user’s face, is encrypted and available only to the Secure Enclave. This data never leaves the device. It’s not sent to Apple, nor is it included in device backups. The following Face ID data is saved, encrypted only for use by the Secure Enclave, during normal operation:

- The mathematical representations of a user’s face calculated during enrollment
- The mathematical representations of a user’s face calculated during some unlock attempts if Face ID deems them useful to augment future matching

Face images captured during normal operation aren’t saved but are instead immediately discarded after the mathematical representation is calculated for either enrollment in Face ID or comparison with enrolled Face ID data.

Exhibit 35, at 29.

Ex. 21 - Infringement Contention Claim Chart
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	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device's neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36. Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It's designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping). Exhibit 36.
[1i] output a signal comprising information related to the classification of the object.	The Accused Apple Products contain a processor configured to output a signal comprising information related to the classification of the object. For example, Face ID on the Apple iPhone 17 unlocks the phone upon classifying, and recognizing, the individual in the image.

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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22

Ex. 21 - Infringement Contention Claim Chart
U.S. Patent No. 12,298,394 (“the ’394 patent”)

	About Face ID advanced technology Learn how Face ID helps protect your information on your iPhone and iPad Pro. Much of our digital lives are stored on iPhone and iPad, and it's important to protect that information. In the same way that Touch ID revolutionized authentication using a fingerprint, Face ID revolutionizes authentication using facial recognition. Face ID provides intuitive and secure authentication enabled by the state-of-the-art TrueDepth camera system with advanced technologies to accurately map the geometry of your face. With a simple glance, Face ID securely unlocks your iPhone or iPad Pro. You can use it to authorize payments with Apple Pay, purchases from the App Store, Book Store, and other Apple media services, and more. Developers can also allow you to use Face ID to sign into their apps. Exhibit 36.
Claim 15	Accused Apple Products
[15pre] A system, comprising:	Each Accused Apple Product constitutes a system. <i>See supra</i> analysis for limitation [1pre].
[15a] a structured light projector configured to emit a plurality of light beams, arranged in a first spatial arrangement relative to each other, towards an objection;	The Accused Apple Products contain a structured light projector configured to emit a plurality of light beams, arranged in a first spatial arrangement relative to each other, towards an object. <i>See supra</i> analysis for limitation [1a].
[15b] a semiconductor detector configured to receive a light reflective from the object and output an image characterizing an intensity of the reflective light; and	The Accused Apple Products contain a semiconductor detector configured to receive a light reflective from the object and output an image characterizing an intensity of the reflective light. <i>See supra</i> analysis for limitation [1b].

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[15c] a processor configured to:	The Accused Apple Products contain a processor. <i>See supra</i> analysis for limitation [1c].
[15d] select at least one reflection region of the image that characterizes the reflective light;	The Accused Apple Products contain a processor configured to select at least one reflection region of the image that characterizes the reflective light. <i>See supra</i> analysis for limitation [1d].
[15e] determine a corresponding reflection region in reference image data derived from a plurality of reference images;	The Accused Apple Products contain a processor configured to determine a corresponding reflection region in reference image data derived from a plurality of reference images. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispooofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22. “Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users.” Exhibit 35, at 29.

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	“An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.
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Ex. 21 - Infringement Contention Claim Chart
U.S. Patent No. 12,298,394 (“the ’394 patent”)

Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

Ex. 21 - Infringement Contention Claim Chart
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Biometric security

Passcodes and passwords are essential to the security of Apple devices. At the same time, users require convenient access to their devices, often more than a hundred times a day. Biometric authentication provides a way to retain the security of a strong passcode—or even strengthen the passcode or password because it need not be entered manually—while providing the convenience of swiftly unlocking with a finger press or glance. Optic ID, Face ID, and Touch ID don’t replace a passcode or password, but in most situations they do make access faster and easier.

Apple’s biometric security architecture relies on a strict separation of responsibilities between the biometric sensor and the Secure Enclave, and a secure connection between the two. The sensor captures the biometric image and securely transmits it to the Secure Enclave. During enrollment, the Secure Enclave processes, encrypts, and stores the corresponding Optic ID, Face ID, and Touch ID template data. During matching, the Secure Enclave compares incoming data from the biometric sensor against the stored templates to determine whether to unlock the device or respond that a match is valid (for Apple Pay, in-app, and other uses of Optic ID, Face ID, and Touch ID). The architecture supports devices that include both the sensor and Secure Enclave (such as iPhone, iPad, many Mac computers, and Apple Vision Pro), as well as the ability to physically separate the sensor into a peripheral that’s then securely paired to the Secure Enclave in a Mac with Apple silicon.

Exhibit 35, at 20.

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Face ID security

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Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

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Facial matching security

Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later. Furthermore, it’s designed to work indoors, outdoors, and even in total darkness. An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks. Face ID data, including mathematical representations of a user’s face, is encrypted and available only to the Secure Enclave. This data never leaves the device. It’s not sent to Apple, nor is it included in device backups. The following Face ID data is saved, encrypted only for use by the Secure Enclave, during normal operation:

- The mathematical representations of a user’s face calculated during enrollment
- The mathematical representations of a user’s face calculated during some unlock attempts if Face ID deems them useful to augment future matching

Face images captured during normal operation aren’t saved but are instead immediately discarded after the mathematical representation is calculated for either enrollment in Face ID or comparison with enrolled Face ID data.

Exhibit 35, at 29.

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	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device’s neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36. Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It’s designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping). Exhibit 36.
[15f] determine a distribution of light intensity of the reflection region of the image;	The Accused Apple Products determine a distribution of light intensity of the reflection region of the image. <i>See supra</i> analysis for limitations [1e] and [1f].
[15g] compare the distribution of light intensity of the reflection region of the image to a distribution of light intensity of the corresponding reflection region in the reference image data;	The Accused Apple Products contain a processor configured to compare the distribution of light intensity of the reflection region of the image to a distribution of light intensity of the corresponding reflection region in the reference image data. <i>See supra</i> analysis for limitation [1g].

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[15h] determine a classification of the object based on the distribution of light intensity comparison of the reflection region of the image and the corresponding reflection region of the reference images;	The Accused Apple Products determine a classification of the object based on the distribution of light intensity comparison of the reflection region of the image and the corresponding reflection region of the reference images. <i>See supra</i> analysis for limitation [1h].
[15i] output a signal comprising information related to the classification of the object.	The Accused Apple Products contain a processor configured to output a signal comprising information related to the classification of the object. <i>See supra</i> analysis for limitation [1i].